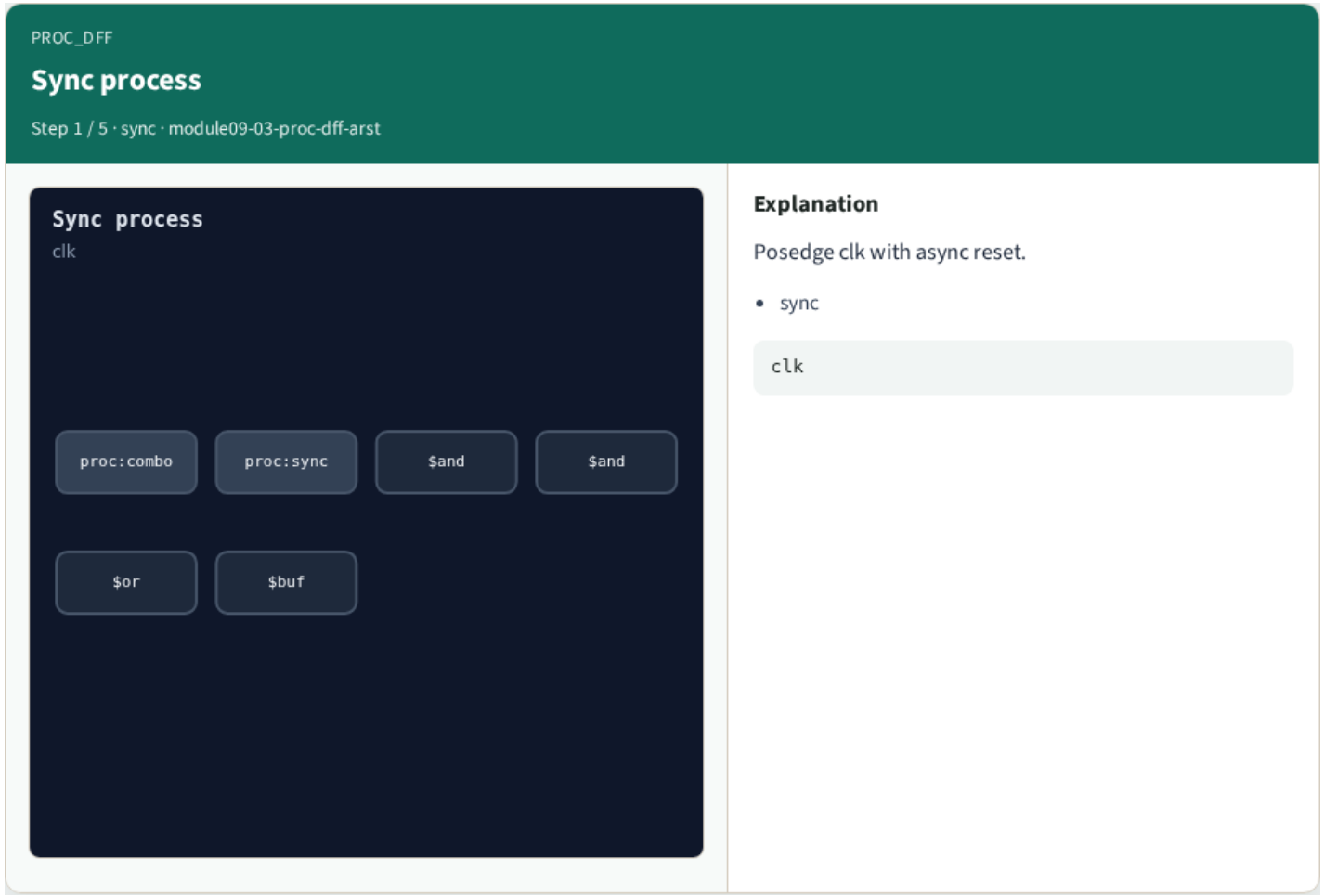
Why proc dff

Sync processes become flip-flops; async reset rides on the ARST ports

The idea

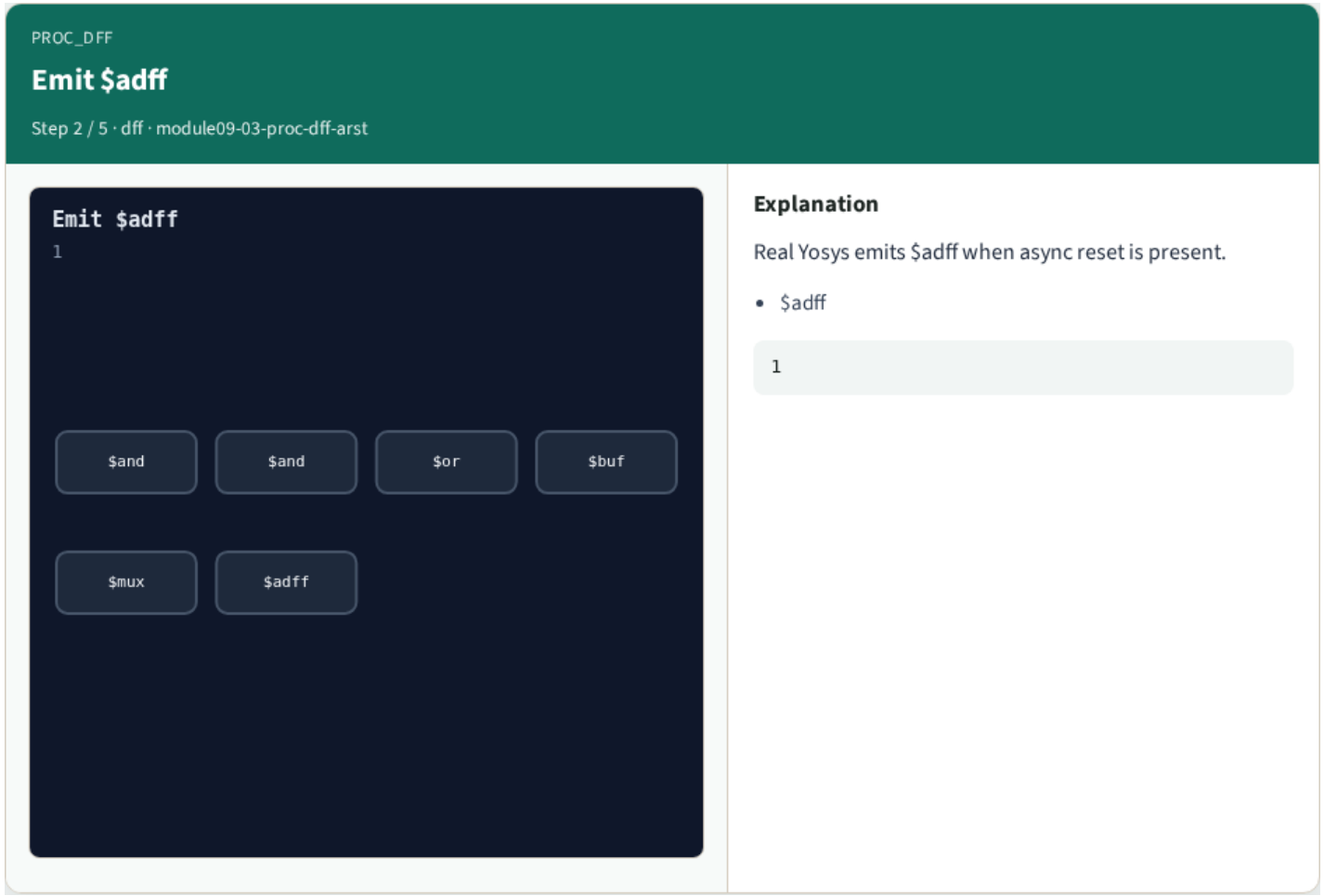
- GOLDEN: one dollar adff with arst; processes left = 0
- Real Yosys binds the same

Sync process



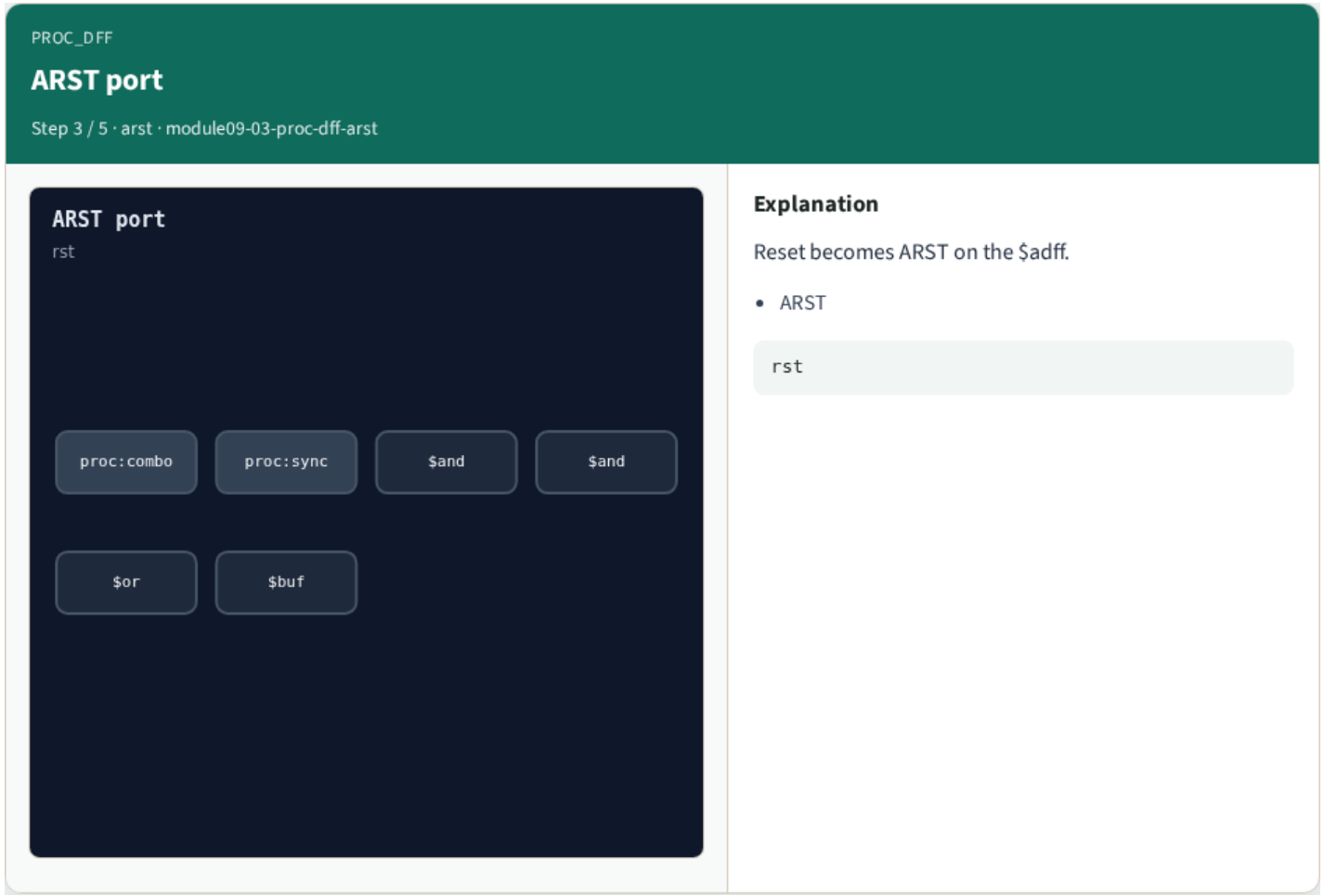
Sync process

Emit $adff



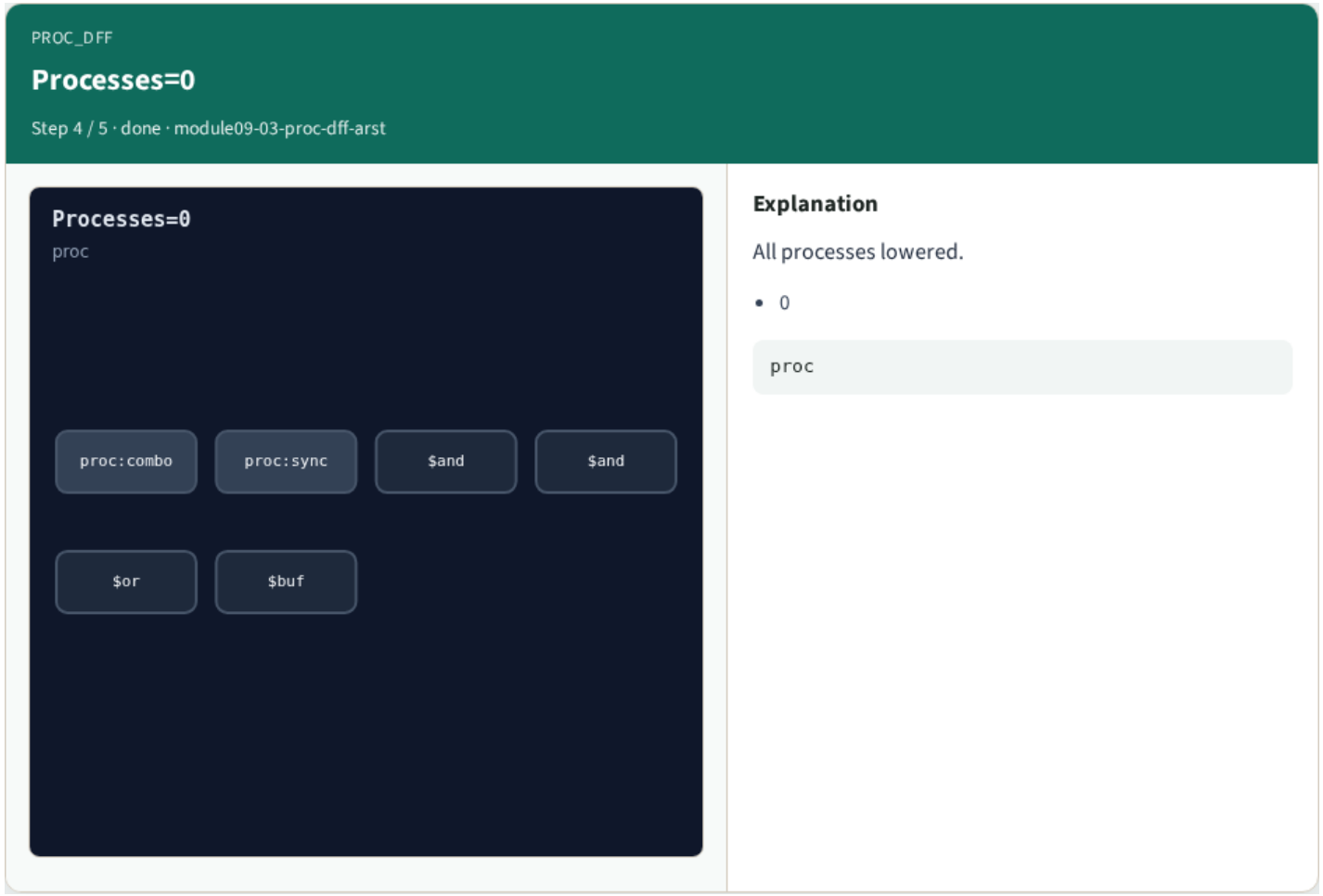
Emit $adff

ARST port



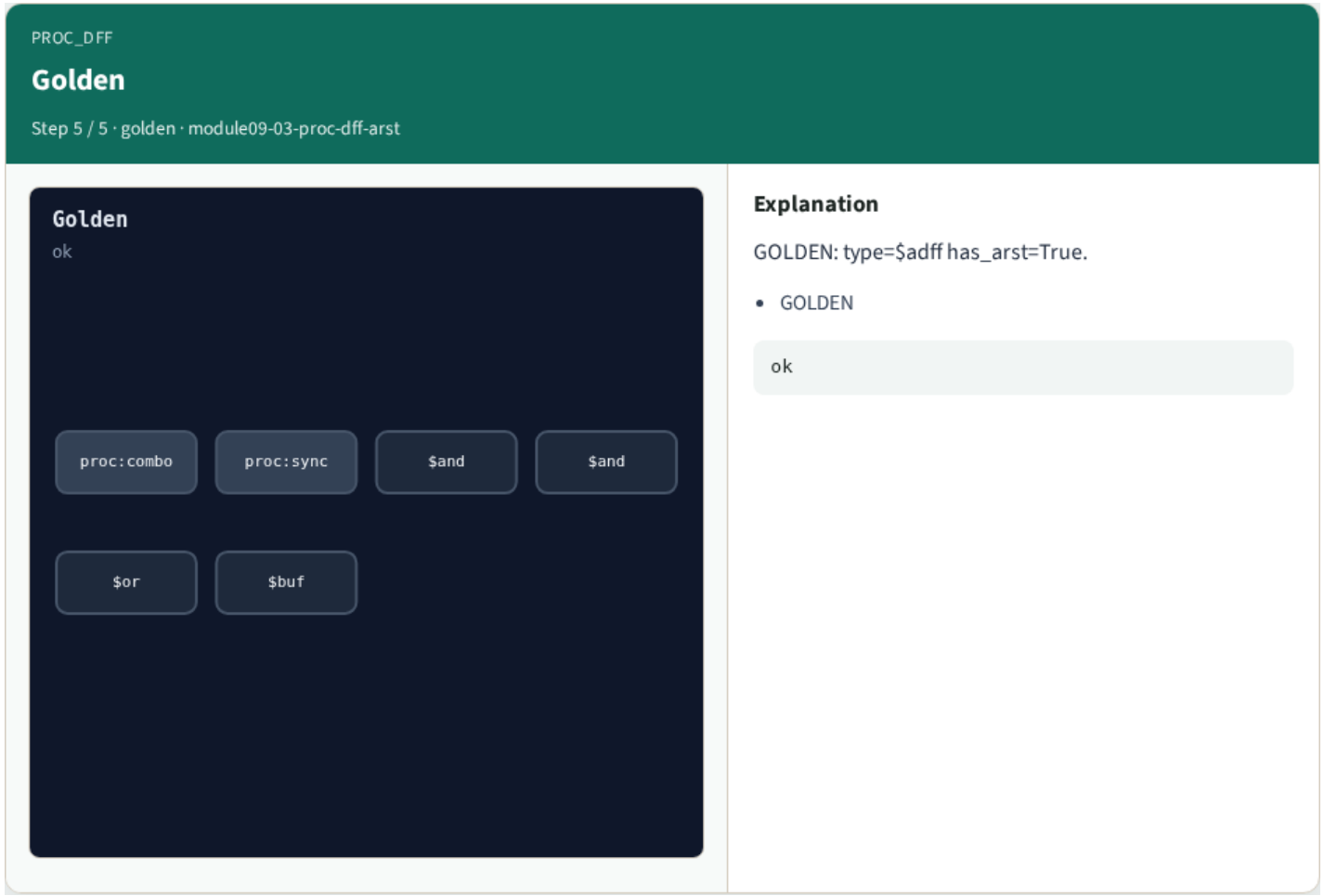
ARST port

Processes=0



Processes=0

Golden



Golden

Browser lab track

- Open **proc-dff-arst**
- Emit the $adff with reset ports, clear all processes, and confirm the real bind

Implement track

- Call algo_proc_dff(); assert reg_type is $adff, has_arst is True, processes == 0
- Then run partc_bind.py

Pitfalls

- Emitting plain $dff when async reset is present
- Stopping before processes reach zero

Try these

- From the common folder

Try these — try these

```
cd courses/learn_yosys/common
python3 algo_solvers.py
python3 partc_bind.py
```

Pseudocode

- Here is the check in plain steps

Pseudocode — try these

```
run proc_mux first
for each sync process:
  if reset present: emit $adff (CLK,D,Q,ARST)
  else: emit $dff (CLK,D,Q)
ok if reg_type==$adff and processes==0
```

Next

- Complete the quiz for this part
- Next: opt_clean and opt_expr goldens